Harsha Srinatth Ryali

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Professional Summary

Computer Science and Business Systems undergraduate with strong foundations in Full Stack Web Development, Data Structures and Algorithms, and Machine Learning Fundamentals. Experienced in building scalable MERN-stack applications using React.js, Node.js, Express.js, MongoDB, JWT Authentication, and Socket.IO. Passionate about developing impactful software solutions that combine modern web technologies with data-driven insights.

Education

SRKR Engineering College, Bhimavaram
B.Tech in Computer Science and Business Systems
CGPA: 8.75 / 10.00

Sri Chaitanya Junior College, Andhra Pradesh
Pre-University Course in Physics, Chemistry, Mathematics (MPC)
GPA: 8.84 / 10.00

Projects

Centralized Academic & Skill Management Platform (MERN + ML) *student-frontend-phi.vercel.app*

Tech Stack: React.js, Node.js, Express.js, MongoDB, Socket.IO, JWT, Python, Machine Learning

- Engineered a scalable full-stack platform with Student, Faculty, and HOD portals to centralize academic tracking, attendance, achievements, and institutional workflows.
- Designed and implemented JWT-based authentication and role-based authorization, enabling secure access control and faculty-driven verification processes.
- Built real-time communication features using Socket.IO, including notifications, threaded doubt discussions, and activity updates to improve user engagement.
- Developed a Skill Exchange module supporting peer-to-peer learning through course creation, collaborative knowledge sharing, and structured feedback.
- Integrated machine learning models through Python APIs to predict student outcomes, placement readiness tiers, and participation trends using academic and engagement data.
- Designed scalable REST APIs and modular backend architecture to support future expansion and maintainability.

SalonFlow – Real-Time Salon Queue & Booking Management SaaS *salonflow-eta.vercel.app*

Tech Stack: React.js, Redux Toolkit, Tailwind CSS, Node.js, Express.js, PostgreSQL, Drizzle ORM, Socket.IO, Redis, Firebase Auth, Stripe, Docker

- Built a multi-role SaaS platform with dedicated Admin, Receptionist, Employee, and Customer portals to digitize end-to-end salon operations.
- Designed a real-time live-queue engine (Socket.IO + Redis adapter) using historical service-time sampling to deliver confidence-based wait-time estimates and proactive turn-alerts.
- Implemented secure authentication combining JWT sessions with Firebase Phone OTP staff verification, hardened with rate limiting, login lockout, and audit logging via Helmet and express-rate-limit.
- Developed booking, loyalty, membership, and offers modules on PostgreSQL with Drizzle ORM, plus Cloudinary media uploads and PDFKit-generated receipts and invoices.
- Integrated Stripe subscription billing with webhook-driven Basic/Pro plan management, and containerized the frontend/backend as independent services with Docker Compose.

Technical Skills

Programming Languages: JavaScript, Python, Java, C++, SQL

Frontend: React.js, Next.js, HTML5, CSS3, Redux, Context API

Backend: Node.js, Express.js, REST APIs, JWT Authentication, Socket.IO

Databases & Tools: MongoDB, PostgreSQL, Redis, Git, Docker

Core Concepts: Data Structures and Algorithms, System Design Basics, Machine Learning Fundamentals

Achievements & Certifications

- Shortlisted at the college-level hackathon among 300+ competing teams for presenting an innovative software solution.
- Recognized for designing and implementing an end-to-end scalable full-stack platform demonstrating strong problem-solving and software engineering skills.